



DEPARTMENT OF NUCLEAR MEDICINE

18-F- FDG PET-CT SCAN

Name	Mrs. Anasuya	Date of study	09-04-2025
Age / Sex	77 Years / Female	Hospital No	1221802
Referred by	Dr. Suresh Balaji MCh	Scan Number	2504910 PET

Clinical indication:- Case of CUPS. CT (25-03-2025) - Moderate to gross ascites. Endoscopy (28-03-2025) - Pangastritis. HPE (31-03-2025) - Sigmoid colon - Suspicious of poorly differentiated malignancy - ? primary ? secondary. HPE (31-03-2025) - Body of stomach - Poorly differentiated malignancy. IHC (03-04-2025)- Body of stomach mucosa - Features of poorly cohesive carcinoma with signet ring cells. CA 125 (31-03-2025) - 937U/mL.

Procedure:- After 6 hours of fasting, serum blood glucose levels were checked (96 mg/dl). Whole body images (skull to mid thigh) were acquired 60 minutes after intravenous injection of 18F-FDG. **Intravenous contrast was used.** Plain water was used as oral contrast. Images were reconstructed in transaxial, coronal and sagittal views. Standardized uptake values normalized to lean body mass were calculated. Serum Creatinine – 0.9 mg/dl. Salient anatomical findings are mentioned.

Findings:-

Head & Neck :-

Physiological uptake of FDG is noted in both cerebral hemispheres, basal ganglia, thalami and both cerebellar hemispheres. Brain parenchyma appears morphologically normal.

No abnormal FDG uptake / structural abnormality is noted in the posterior wall of the nasopharynx, fossa of Rosenmuller and retromolar trigone. No obvious morphological abnormality is noted in the salivary glands, oropharynx and the larynx.

Heterogeneous nodules are noted in both lobes of the thyroid gland, largest measuring 20 x 15 mm in size - USG / FNAC correlation may be considered.

Increased FDG uptake (SUV Max 2.84) is noted in few prominent right cervical level II lymph nodes, largest measuring 8 x 9 mm in size.

Chest:-

No significant FDG uptake is noted in few enhancing nodules in the right breast parenchyma, most evident in the retro-areolar region, measuring 5.5 mm in size - ? nature.

Mildly increased FDG uptake (SUV Max 2.0) is noted in the enlarged and enhancing right axillary lymph nodes, largest measuring 5.5 x 15 mm in size.

No significant FDG uptake is noted in few enhancing areas in the left breast parenchyma, without any significant FDG uptake - ? significance.

No significant FDG uptake (SUV Max 0.83) is noted in the visualized left axillary lymph nodes, largest



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measuring 4 mm in size.

Thoracic esophagus appears normal without any obvious FDG avidity.

Tiny sub-solid nodules are noted in both lungs, without any significant FDG uptake, largest is in the superior segment of the right lung lower lobe, measuring 3.5 mm in size.

Increased FDG uptake is noted in the following lymph nodes (size of the largest node is mentioned) :

- a) Subcarinal, peri-bronchial bilaterally (SUV Max 3.70), measuring < 1 cm
- b) Right upper paratracheal (SUV Max 1.94), largest measuring 5 mm

Minimal left pleural effusion is noted

Abdomen, Pelvis and inguinal regions :-

Homogeneously increased FDG uptake (SUV Max 2.59) is noted in diffusely increased contrast enhancement and wall thickening in the entire body and antro-pyloric region of the stomach, with involvement of the cardia of the stomach. Maximum single wall thickness is approximately 14 mm in size.

Variably increased FDG uptake is noted in the following lymph nodes (size of the largest node is mentioned) :

- a) Periportal / portacaval (SUV Max 2.51), measuring 6 x 11 mm
- b) Perigastric lymph nodes adjoining the greater curvature of the stomach / omental nodules (SUV Max 0.78), largest measuring 6.5 x 10.5 mm
- c) Retroperitoneal lymph nodes (SUV Max 1.41), largest measuring 3 mm
- d) Few mesenteric lymph nodes (SUV Max 1.34), largest measuring 6 x 9 mm

Perigastric stranding is noted with few dilated venous channels in the perigastric region.

Thickening of the soft tissues of bilateral paracolic regions, bilateral pararenal fascia is noted, with mildly increased FDG uptake (SUV Max 1.59). Irregular soft tissue stranding is noted in the perisplenic region and omentum, with minimal FDG uptake (SUV Max 1.05).

No abnormal FDG uptake is noted in the right adnexa. No significant FDG uptake is noted in the left adnexa, which measures approximately 13 x 26.5 mm in size.

Moderate to gross ascites is noted.

Increased FDG uptake (SUV Max 3.24) is noted in heterogeneously enhancing lesion replacing the right adrenal gland, measuring 10 x 25 mm in size.

Increased FDG uptake (SUV Max 4.72) is noted in the enlarged and bulky left adrenal gland, with heterogeneous contrast enhancement measuring 12 x 21 mm in size.



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Liver shows physiological FDG uptake. Liver measures 13.1 cm craniocaudally. No obvious focal lesions are noted in the liver. Liver shows homogenous parenchymal tissue density. There is no evidence of intrahepatic biliary dilatation.

Minimal diffuse enhancement is noted in the gall bladder wall, with a maximum thickness of approximately 3.5 mm, without any obvious FDG uptake.

Portal vein is well opacified without any obvious filling defects noted within.

Spleen shows physiological FDG uptake. Spleen measures 8.9 cm craniocaudally. No obvious focal lesions are noted in the spleen.

No abnormal FDG uptake in the pancreas. The pancreas has a normal size and configuration.

Bilateral minimal perinephric stranding is noted.

Physiological FDG uptake is noted in rest of the visualized bowel loops.

Tracer excretion is noted into the urinary bladder. No definite FDG avid lesion is noted in the urinary bladder. No FDG avid lymph nodes are noted in both inguinal region.

Musculo-skeletal system :-

Minimal thickening of the subcutaneous tissues of visualized part of the thighs and gluteal region is noted.

Foci of intense FDG uptake (SUV Max 10.98) are noted in the following bones :

- a) Both femurs
- b) Multiple focal lesions in the pelvic bones on both sides
- c) Sacrum
- d) Sternum
- e) Multiple ribs on both sides
- f) Both scapula
- g) Both clavicles
- h) Both humerus
- i) Right ramus and condyle of the mandible
- j) C2, C3 and C4 vertebrae
- k) D5, D6, D7, D8, D9, D10, D11 and D12 vertebrae
- l) L1, L2, L3, L4 and L5 vertebrae

Heterogeneously increased FDG uptake (SUV Max 2.10) is noted in rest of the visualized bones.

Extensive tiny sclerotic lesions are noted throughout the visualized bones.



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IMPRESSION:-

In a known case of CUPS / PET for evaluation of disease status :

- 1. Variably increased metabolic activity in diffuse wall thickening and contrast enhancement involving the entire body of the stomach as described - malignant lesion.**
- 2. Upper abdominal lymph nodal metastasis as described. Omental spread as described.**
- 3. Thickening of the bilateral paracolic regions, bilateral pararenal fascia and perisplenic soft tissues as described - disease dissemination.**
- 4. Ascites as described. Bilateral adrenal metastases as described.**
- 5. Minimal left pleural effusion as described. Mediastinal lymph nodes with increased metabolic activity - ? Metastases.**
- 6. Extensive bones / bone marrow metastasis as described.**

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